

02427 Advanced Time Series Analysis

Computer exercise 1

This exercise starts with some modelling with non-linear models and afterwards methods for nonparametric estimation are used to identify functional dependencies.

Part 1

Simulate different non-linear models. Three good models to start out with are:

- SETAR(2;1;1),
- IGAR(2;1), and
- MMAR(2;1).

Try different parameters.

For reporting, write the models and discuss the most essential features of each model together with some informative plots.

Hints (R and Matlab) Check out the script `3dPlotting.R`. It starts out showing a simple way to implement a simulation of a process with a for-loop.

Part 2

Compute the theoretical conditional mean,

$$M(x) = E\{X_{t+1} \mid X_t = x\},$$

for a SETAR(2,1,1) of your own choice.

Simulate 1000 values from the chosen SETAR model. Use these simulated data and a local regression model to estimate the conditional mean

$$\hat{M}(x) = E\{X_{t+1} \mid X_t = x\}.$$

Try different bandwidths and comment on your findings.

Hints (R) You can get inspired by the script in `3dPlotting.R` and explore local regression with `loess()` or `lm()`. See `?loess` for which type of kernel etc. is used. Using `lm()` and doing the weights with your own kernel function enables more models to be fitted, which is especially useful for later computer exercises.

Hints (Matlab) The function in `regsmooth1D.m` does one-dimensional local polynomial regression.

Part 3

Choose a model where the conditional variance of X_{t+1} depends nonlinearly on X_t and estimate the conditional variance

$$\hat{V}(x) = Var\{X_{t+1} \mid X_t = x\}.$$

Use a leave-one-out approach to determine the best bandwidth for a kernel of your choice.

Part 4

During the heating season, the heat-loss coefficient of buildings, U_a , is often estimated under the assumption that the internal temperature is kept constant, so that the heat load is described by

$$\Phi_t = U_a(T_t^i - T_t^e) + \epsilon_t,$$

where Φ_t is heat load, U_a is the heat loss coefficient and T_t^i and T_t^e are the internal and external temperatures. The problem with this approach is that the heat-loss coefficient depends nonlinearly on the wind speed. Using the

data from `DataPart4.csv`, model U_a as a non-parametric function of the wind speed, i.e.

$$\tau_t = U_a(W_t) (T_t^i - T_t^e) + \epsilon_t.$$

Plot the estimated function $U_a(W_t)$.

Hints It might be useful to rewrite the deterministic part of the model as

$$U_a(W_t) = \frac{\tau_t}{T_t^i - T_t^e}.$$

R In `3dPlotting.R` local polynomial regression is carried out, both using `loess()` and `lm()`. Conditional parametric models can be fitted with either:

- using `loess()` the parameter `parametric` needs to be altered;
- using `lm()` the way the weights are calculated needs to be altered.

Matlab The function in `regsmooth2D.m` fits either a local polynomial regression model or a conditional parametric model. Look into it to learn what is done differently for the two modelling techniques.

Part 5

Open the data from `DataPart5.csv` and model it using an ARMA model, increasing the model order assisted by the ACF and PACF, until the residuals look white. When you are reasonably satisfied compute the LDF of the residuals. Do you find any significant non-linearities? Propose a better model structure.

Hints Check the n -step residuals versus residuals plot, where n is the significant LDF lag, i.e., e_t versus e_{t-n} .

R The script `ldf.R` is a way to estimate lagged dependent functions.

Matlab The script in `ldf.m` does an estimation of Lagged Dependent Functions. The script in `ldfone.m` can be used to see the estimated dependence function for some lag.